Que :- To study printer deadlock scenarios, analyze resource contention, and propose synchronization strategies ensuring smooth, efficient printing operations.

Code:

v

```html
code.html
code.html > html > head > style > *
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>OS Deadlock Simulation</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            min-height: 100vh;
            color: #333;
        }

        .navbar {
            background: rgba(255, 255, 255, 0.95);
            backdrop-filter: blur(10px);
            padding: 1rem 2rem;
            box-shadow: 0 2px 20px rgba(0,0,0,0.1);
            position: fixed;
            width: 100%;
            top: 0;
            z-index: 1000;
        }

        .nav-container {
            max-width: 1200px;
            margin: 0 auto;
            display: flex;
            justify-content: space-between;
            align-items: center;
        }

        .logo {
            font-size: 1.5rem;
            font-weight: bold;
            color: #667eea;
        }

        .nav-links {
            display: flex;
            list-style: none;
            gap: 2rem;
        }

        .nav-links a {
            text-decoration: none;
            color: #333;
            font-weight: 500;
            padding: 0.5rem 1rem;
            border-radius: 25px;
            transition: all 0.3s ease;
        }

        .nav-links a:hover,
        .nav-links a.active {
            background: #667eea;
            color: white;
        }

        .container {
            max-width: 1200px;
            margin: 0 auto;
            padding: 120px 2rem 2rem;
            min-height: 100vh;
```

```html
code.html
code.html > html > head > style > *
<html lang="en">
<head>
    <style>
        }

        /* Start Page */
        .start-page {
            text-align: center;
            padding-top: 4rem;
        }

        .start-title {
            font-size: 4rem;
            color: white;
            margin-bottom: 2rem;
            text-shadow: 2px 2px 4px rgba(0,0,0,0.3);
        }

        .start-subtitle {
            font-size: 1.5rem;
            color: rgba(255,255,255,0.9);
            margin-bottom: 4rem;
            max-width: 600px;
            margin-left: auto;
            margin-right: auto;
        }

        .start-btn {
            background: #ff6b6b;
            color: white;
            border: none;
            padding: 1.5rem 3rem;
            font-size: 1.5rem;
            border-radius: 50px;
            cursor: pointer;
            transition: all 0.3s ease;
            box-shadow: 0 10px 30px rgba(255,107,107,0.4);
            text-transform: uppercase;
            letter-spacing: 2px;
        }

        .start-btn:hover {
            transform: translateY(-5px);
            box-shadow: 0 15px 40px rgba(255,107,107,0.6);
            background: #ff5252;
        }

        /* Explanation Page */
        .explanation-page {
            background: white;
            border-radius: 20px;
            padding: 3rem;
            box-shadow: 0 20px 60px rgba(0,0,0,0.1);
            margin-bottom: 2rem;
        }

        .section-title {
            font-size: 2.5rem;
            color: #667eea;
            margin-bottom: 1.5rem;
            text-align: center;
        }

        .explanation-content {
            line-height: 1.8;
            font-size: 1.1rem;
            color: #555;
        }

        .conditions-list {
            display: grid;
```

```html
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            color: white;
            border: none;
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        }

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            font-size: 2.5rem;
            color: #667eea;
            margin-bottom: 1.5rem;
            text-align: center;
        }

        .explanation-content {
            line-height: 1.8;
            font-size: 1.1rem;
            color: #555;
        }

        .conditions-list {
            display: grid;
```

```html
code.html
code.html > html > head > style > .process
<html lang="en">
<head>
    <style>
        .resource {
            transition: all 0.3s ease;
        }

        .resource.held {
            border-color: #ff6b6b;
            background: #ffe6e6;
        }

        .resource.requested {
            border-color: #feca57;
            background: #fff8e1;
        }

        .resource-name {
            font-size: 1.5rem;
            font-weight: bold;
            margin-bottom: 1rem;
        }

        .process-status {
            display: grid;
            grid-template-columns: repeat(4, 1fr);
            gap: 1rem;
            margin-top: 2rem;
        }

        .process {
            padding: 1.5rem;
            border-radius: 10px;
            text-align: center;
            font-weight: bold;
        }

        .process.running { background: #4ecdc4; color: white; }
        .process.waiting { background: #ff6b6b; color: white; }
        .process.deadlocked { background: #95a5a6; color: white; }

        .controls {
            text-align: center;
            margin: 2rem 0;
        }

        .control-btn {
            background: #667eea;
            color: white;
            border: none;
            padding: 1rem 2rem;
            margin: 0 1rem;
            border-radius: 25px;
            cursor: pointer;
            font-size: 1rem;
            transition: all 0.3s ease;
        }

        .control-btn:hover {
            background: #5a67d8;
            transform: translateY(-2px);
        }

        .control-btn:disabled {
            background: #bdc3c7;
            cursor: not-allowed;
            transform: none;
        }

        @media (max-width: 768px) {
            .nav-links {
```

code.html

code.html > html > head > style

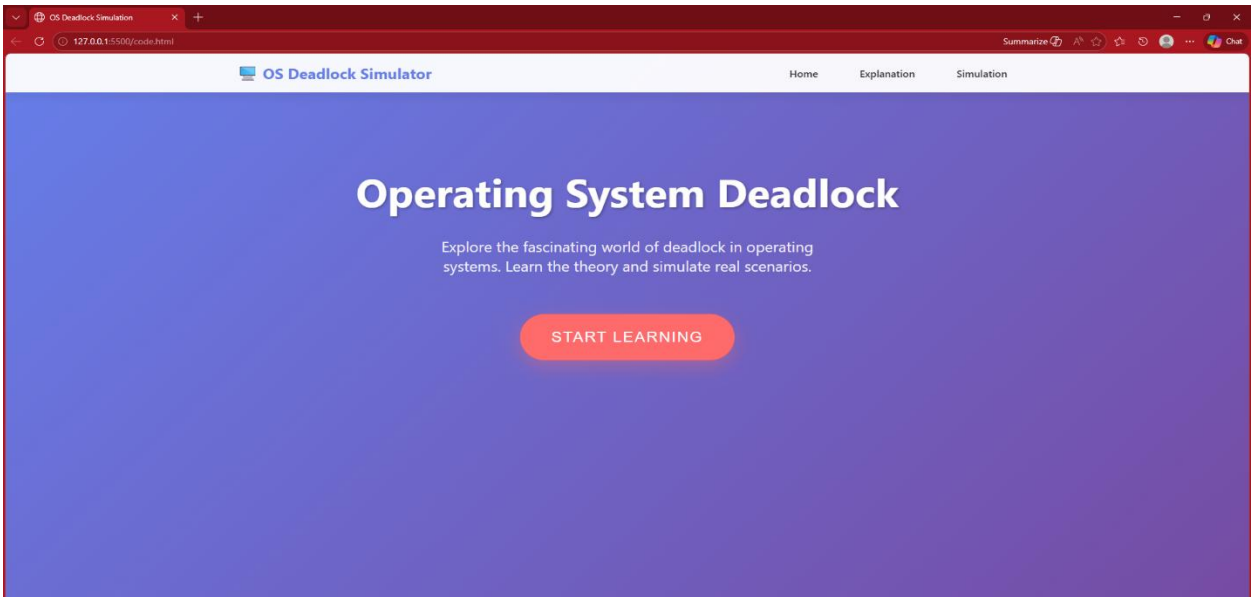
```html
<html lang="en">
<head>
    ...
    </style>
</head>
<body>
    <!-- Navigation Bar -->
    <nav class="navbar">
        <div class="nav-container">
            <div class="logo">🖥 OS Deadlock Simulator</div>
            <ul class="nav-links">
                <li><a href="#" onclick="showPage('start')" class="nav-link">Home</a></li>
                <li><a href="#" onclick="showPage('explanation')" class="nav-link">Explanation</a></li>
                <li><a href="#" onclick="showPage('simulation')" class="nav-link">Simulation</a></li>
            </ul>
        </div>
    </nav>

    <div class="container">
        <!-- Start Page -->
        <div id="start-page" class="start-page">
            <h1 class="start-title">Operating System Deadlock</h1>
            <p class="start-subtitle">Explore the fascinating world of deadlock in operating systems. Learn the theory and simulate real scenarios.</p>
            <button class="start-btn" onclick="showPage('explanation')">Start Learning</button>
        </div>

        <!-- Explanation Page -->
        <div id="explanation-page" class="explanation-page" style="display: none;">
            <h2 class="section-title">What is Deadlock?</h2>
            <div class="explanation-content">
                <p><strong>Deadlock</strong> is a situation where two or more processes are unable to proceed because each is waiting for the other to release a resource. None of the processes can be completed because each process is waiting for a resource

                <h3 style="margin: 2rem 0 1rem 0; color: #667eea;">Coffman's 4 Conditions for Deadlock:</h3>

                <div class="conditions-list">
                    <div class="condition-card">
                        <div class="condition-title">🔒 Mutual Exclusion</div>
                        <p>At least one resource must be held in a non-sharable mode. Only one process at a time can use the resource.</p>
                    </div>
                    <div class="condition-card">
                        <div class="condition-title">🔒 Hold and Wait</div>
                        <p>A process holding at least one resource is waiting to acquire additional resources held by other processes.</p>
                    </div>
                    <div class="condition-card">
                        <div class="condition-title">🔒 No Preemption</div>
                        <p>Resources cannot be preempted. A resource can be released only voluntarily by the process holding it.</p>
                    </div>
                    <div class="condition-card">
                        <div class="condition-title">🔒 Circular Wait</div>
                        <p>A set of processes form a circular chain, where each process is waiting for a resource that the next process in the chain holds.</p>
                    </div>
                </div>

                <div style="text-align: center; margin-top: 3rem;">
                    <button class="simulate-btn" onclick="showPage('simulation')">🚀 Run Deadlock Simulation</button>
                </div>
            </div>
        </div>

        <!-- Simulation Page -->
        <div id="simulation-page" class="simulation-container" style="display: none;">
            <h2 class="section-title">🖥 Deadlock Simulation</h2>
            <div class="explanation-content">
                <p>Watch how 4 processes compete for 3 resources, leading to a classic deadlock scenario!</p>

                <div class="resource-grid">
                    <div class="resource" id="resource-A">
                        <div class="resource-name">Resource A</div>
                        <div>📄 File A</div>
                    </div>
                    <div class="resource" id="resource-B">
                        <div class="resource-name">Resource B</div>
```

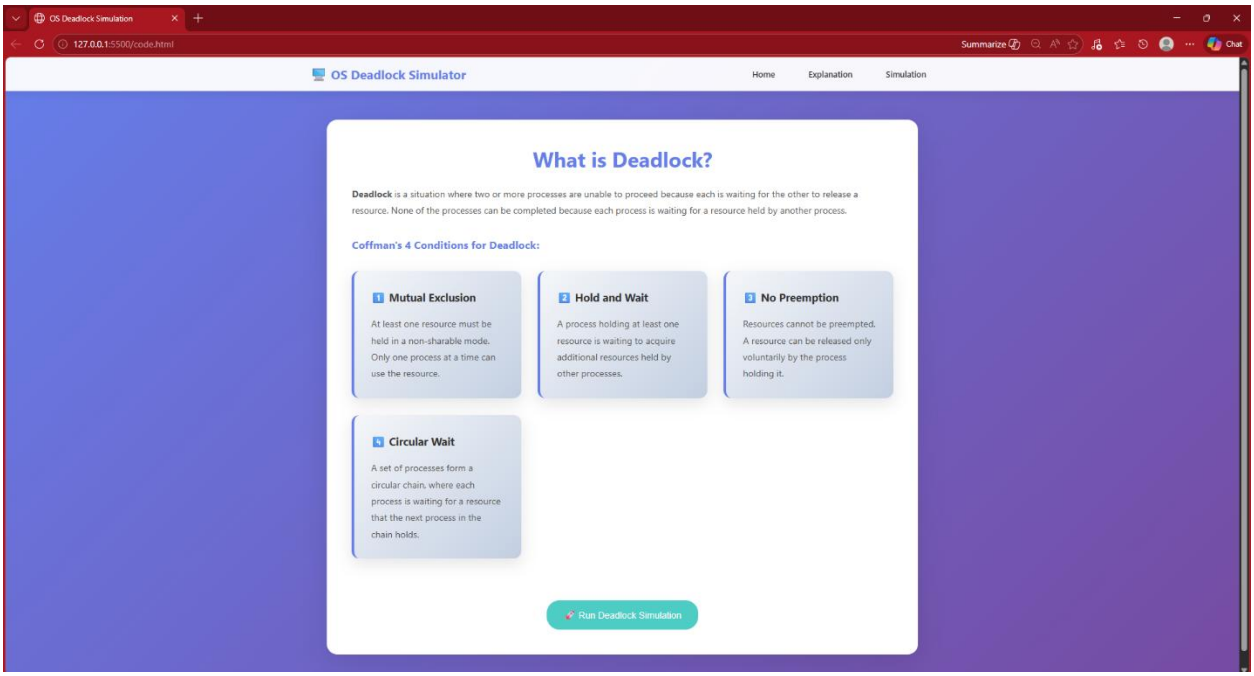
code.html

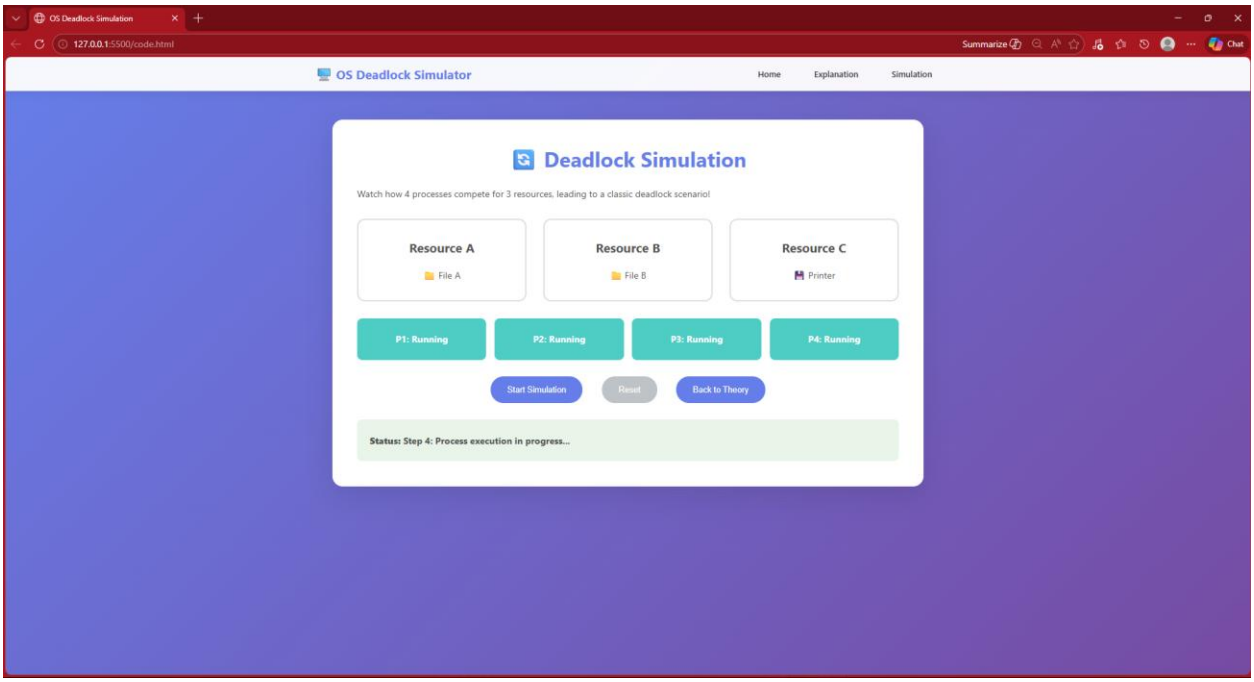
code.html > html > body > script > showPage

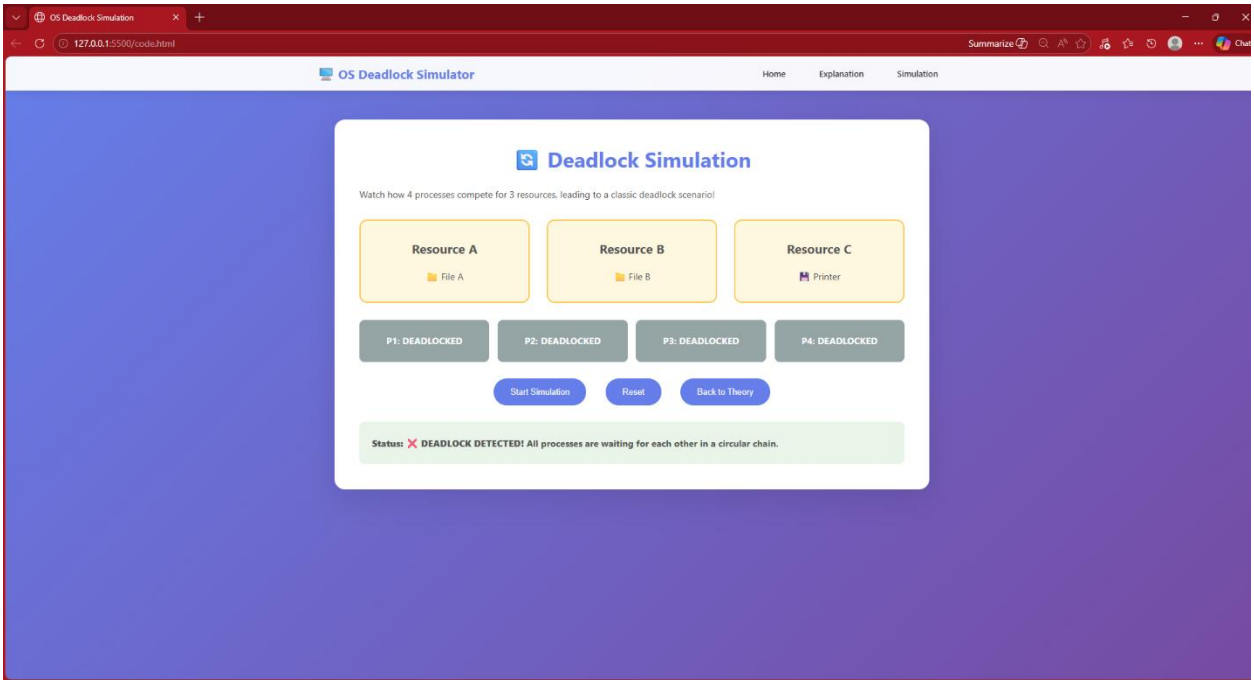
```html
<html lang="en">
<body>
    <script>
        function nextStep() {
            ...
        }

        function updateSimulation(step) {
            // Update resources
            ['A', 'B', 'C'].forEach(res => {
                const el = document.getElementById(`resource-${res}`);
                el.className = 'resource';
                if (step[`resource${res}`]) {
                    el.classList.add(step[`resource${res}`]);
                }
            });

            // Update processes
            ['P1', 'P2', 'P3', 'P4'].forEach((p, i) => {
                const el = document.getElementById(`process-${p}`);
                el.className = `process running`;
                el.textContent = `${p}: Running`;
            });

            document.getElementById('simulation-status').textContent =
                `Step ${simulationStep}: Process execution in progress...`;
        }

        function showDeadlock() {
            // Show deadlock state
            document.getElementById('resource-A').className = 'resource held requested';
            document.getElementById('resource-B').className = 'resource held requested';
            document.getElementById('resource-C').className = 'resource held requested';

            ['P1', 'P2', 'P3', 'P4'].forEach(p => {
                const el = document.getElementById(`process-${p}`);
                el.className = 'process deadlocked';
                el.textContent = `${p}: DEADLOCKED`;
            });

            document.getElementById('simulation-status').innerHTML =
                '❌ <strong>DEADLOCK DETECTED!</strong> All processes are waiting for each other in a circular chain.';

            document.getElementById('reset-btn').disabled = false;
        }

        function resetSimulation() {
            simulationStep = 0;
            ['A', 'B', 'C'].forEach(res => {
                document.getElementById(`resource-${res}`).className = 'resource';
            });

            ['P1', 'P2', 'P3', 'P4'].forEach(p => {
                const el = document.getElementById(`process-${p}`);
                el.className = 'process running';
                el.textContent = `${p}: Running`;
            });

            document.getElementById('simulation-status').textContent = 'Ready to simulate deadlock';
            document.getElementById('reset-btn').disabled = true;
        }

        // Initialize navigation
        document.querySelectorAll('.nav-link').forEach(link => {
            link.addEventListener('click', function(e) {
                e.preventDefault();
            });
        });
    </script>
</body>
</html>
```

Output :


OS Deadlock Simulation
127.0.0.1:5500/code.html
OS Deadlock Simulator
Home
Explanation
Simulation
Operating System Deadlock
Explore the fascinating world of deadlock in operating systems. Learn the theory and simulate real scenarios.
START LEARNING


OS Deadlock Simulation
127.0.0.1:5500/code.html
OS Deadlock Simulator
Home
Explanation
Simulation
What is Deadlock?
Deadlock is a situation where two or more processes are unable to proceed because each is waiting for the other to release a resource. None of the processes can be completed because each process is waiting for a resource held by another process.
Coffman's 4 Conditions for Deadlock:
1 Mutual Exclusion
At least one resource must be held in a non-sharable mode. Only one process at a time can use the resource.
2 Hold and Wait
A process holding at least one resource is waiting to acquire additional resources held by other processes.
3 No Preemption
Resources cannot be preempted. A resource can be released only voluntarily by the process holding it.
4 Circular Wait
A set of processes form a circular chain, where each process is waiting for a resource that the next process in the chain holds.
Run Deadlock Simulation


OS Deadlock Simulation
127.0.0.1:5500/code.html
OS Deadlock Simulator
Home
Explanation
Simulation
Deadlock Simulation
Watch how 4 processes compete for 3 resources, leading to a classic deadlock scenario!
Resource A
File A
Resource B
File B
Resource C
Printer
P1: Running
P2: Running
P3: Running
P4: Running
Start Simulation
Reset
Back to Theory
Status: Step 4: Process execution in progress...


OS Deadlock Simulation
127.0.0.1:5500/code.html
OS Deadlock Simulator
Home
Explanation
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Deadlock Simulation
Watch how 4 processes compete for 3 resources, leading to a classic deadlock scenario!
Resource A
File A
Resource B
File B
Resource C
Printer
P1: DEADLOCKED
P2: DEADLOCKED
P3: DEADLOCKED
P4: DEADLOCKED
Start Simulation
Reset
Back to Theory
Status: ❌ DEADLOCK DETECTED! All processes are waiting for each other in a circular chain.